ML (Maximum Likelihood)

DML $y(t) \sim N(Ax(t), \sigma^2 I)$

$$\min_{\theta, X, \sigma^2} \frac{1}{\sigma^2}\|Y - AX\|_F^2 + ML\log\pi\sigma^2$$

Pf: $p(y) = \left(\frac{1}{\pi\sigma^2}\right)^M e^{-\frac{1}{\sigma^2}(y-Ax)^H(y-Ax)}$

$$\sum_{l=1}^{L}\log p(y_l) = -\frac{1}{\sigma^2}\sum_{l=1}^{L}\|y_l - Ax_l\|_2^2 - ML\log\pi\sigma^2$$

Given θ, est X

Nonlinear LS $\min_{\theta, X}\|Y - A(\theta)X\|_F^2$

$$\hat{X} = A^\dagger(\theta)Y = (A^HA)^{-1}A^HY$$

Given $\theta, \hat{X}$, est σ^2

$$\hat{\sigma}^2 = \frac{1}{M}\mathrm{tr}(\mathrm{Proj}_A^\perp \hat{R})$$

$(\mathrm{Proj}_A^\perp)^2 = \mathrm{Proj}_A^\perp$

Pf: $\|Y - A\hat{X}\|_F^2 = \|\underbrace{(I - A(A^HA)^{-1}A^H)}_{\mathrm{Proj}_A^\perp}Y\|_F^2 = \mathrm{tr}(\mathrm{Proj}_A^\perp YY^H \mathrm{Proj}_A^\perp)$

$= L\,\mathrm{tr}(\mathrm{Proj}_A^\perp \hat{R})$

$\min_{\sigma^2}\frac{1}{\sigma^2}\|Y - A\hat{X}\|_F^2 + ML\log\pi\sigma^2$ $\qquad$ $\min_{\sigma^2}\frac{1}{\sigma^2}\mathrm{tr}(\mathrm{Proj}_A^\perp\hat{R}) + M\log\pi\sigma^2$

$$\frac{\partial}{\partial\sigma^2}(\cdot) = -\frac{1}{(\sigma^2)^2}\mathrm{tr}(\mathrm{Proj}_A^\perp\hat{R}) + \frac{M}{\sigma^2} = 0$$

Given $\hat{X}, \hat{\sigma}^2$ est θ

$\hat{\theta} = \arg\min \mathrm{tr}(\mathrm{Proj}_{A(\theta)}^\perp \hat{R})$ 需扫描一圈，每次选K个

On-grid 也要 $\binom{N}{K}$ 种

SML $y(t) \sim N(0, R)$ $\qquad$ $R = APA^H + \sigma^2 I$

$$\min_R \mathrm{tr}(\hat{R}R^{-1}) + \log|R|$$

Given θ, $\hat{\sigma}^2 = \frac{1}{M-K}\mathrm{tr}(\mathrm{Proj}_{A(\theta)}^\perp\hat{R})$

$\hat{P} = A^\dagger(\theta)(\hat{R} - \hat{\sigma}^2 I)[A^\dagger(\theta)]^H$ $\qquad$ P不一定diag

Given $\hat{X}, \hat{\sigma}^2$ est θ

$$\hat{\theta} = \arg\min|A\hat{P}A^H + \hat{\sigma}^2 I|$$

Pf: On the concentrated SML .. Stoica

可能不满足 $P > 0$，也不diag 但若 $L \to \infty$，以概率1满足 $P > 0$

若 $\mathrm{rank}(P) < K$ 无法处理 $\qquad$ $\hat{R} \to R$, $\hat{P} \to A^\dagger \cdot APA^H \cdot A^\dagger = P$